

Charge & Electric Potential

Electric charge:

Charge is a fundamental and characteristic property of the elementary particles which make up matter. These particles are: protons, electrons, and neutrons. Of these, protons are positively charged, electrons negatively charged and neutrons are neutral. The magnitude of the charge of the proton is the same as those of an electron. There are two kinds of charge, positive and negative.

The smallest amount of charge that has been discovered is that of an electron or of a proton. The magnitude of this charge is 1.602×10^{-19} coulomb.

Conductors and insulators:

Conductors:

Conductor" implies that the outer electrons of the atoms are loosely bound and free to move through the material. Therefore, conductors conduct electrical current very easily because of their free electrons.

In copper, the valence electrons are essentially free and strongly repel each other.

Simply stated, most metals are good electrical conductors, most nonmetals are not. Metals are also generally good heat conductors while nonmetals are not.

Some common conductors are copper, aluminum, gold, and silver.

Insulators:

Insulators are materials that have just the opposite effect on the flow of electrons. They do not let electrons flow very easily from one atom to another. Insulators are materials whose atoms have tightly bound electrons. These electrons are not free to roam around and be shared by neighboring atoms. Therefore, Insulators oppose electrical current and make poor conductors.

Some common insulator materials are glass, plastic, rubber, air, and wood.

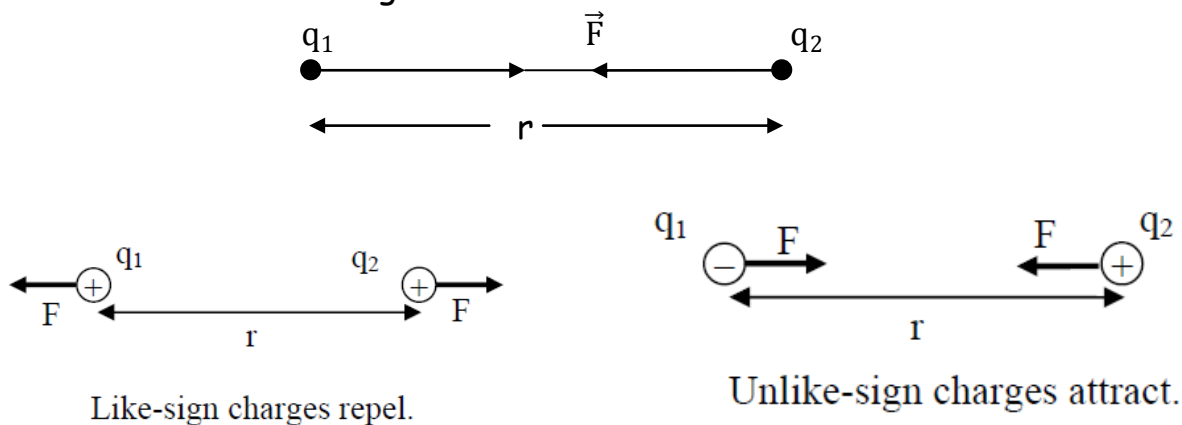
Coulomb's Law:

Statement:

The force of attraction or repulsion between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. The force is along the straight line joining them.

Explanation:

Let us consider two point charges q_1 and q_2 at rest at a distance r from each other as shown in the figure.



Then according to the statement, the mutual force between the two point charges,

$$F \propto q_1 q_2, \text{ when } r \text{ is constant.}$$

$$F \propto \frac{1}{r^2}, \text{ when } q_1 \text{ and } q_2 \text{ are constants.}$$

Now, by the rule of proportionality, we can write,

$$F \propto \frac{q_1 q_2}{r^2}$$
$$\text{or, } F = c \frac{q_1 q_2}{r^2} \dots\dots\dots (1)$$

where c is the constant of proportionality, which is usually written in a more complex way as $\frac{1}{4\pi\epsilon_0}$, where ϵ_0 is the permittivity constant.

Then the Coulomb's law takes the form,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \dots\dots\dots (2)$$

In vector notation equation (2) can be written as,

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^3} \vec{r}.$$

Electric Field:

Electric field is defined as the electric force per unit charge. The direction of the field is taken to be the direction of the force it would exert on a positive test charge. The electric field is radially outward from a positive charge and radially inward toward a negative point charge.

Point Charge
Charged cylinder
Parallel charged plates

Since the measured electric field can depend upon your reference frame, a more general definition of the electric field comes from the Lorentz force law. The electric field can be defined as the electromagnetic force per unit charge in the rest frame of the charge.

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

Electric force *Magnetic force*

Lorentz force law

electric force in Newtons

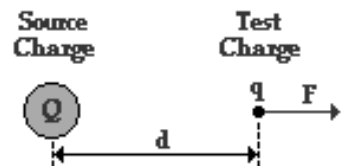
$$\vec{E} = \frac{\vec{F}}{q}$$

charge in Coulombs

A charge that is moving relative to the source will experience part of the force as a magnetic force.

Electric Field Strength:

Electric field strength is a vector quantity; it has both magnitude and direction. The magnitude of the electric field strength is defined in terms of how it is measured. Let's suppose that an electric charge can be denoted by the symbol Q . This electric charge creates an electric field; since Q is the source of the electric field, we will refer to it as the **source charge**. The strength of the source charge's electric field could be measured by any other charge placed somewhere in its surroundings. The charge that is used to measure the electric field strength is referred to as a **test charge** since it is used to *test* the field strength. The test charge has a quantity of charge denoted by the symbol q . When placed within the electric field, the test charge will experience an electric force - either attractive or repulsive. As is usually the case, this force will be denoted by the symbol F . The magnitude of the electric field is simply defined as the force per charge on the test charge.



$$\text{Electric Field Strength} = \frac{\text{Force}}{\text{Charge}}$$

If the electric field strength is denoted by the symbol E , then the equation can be rewritten in symbolic form as

$$E = \frac{F}{q}$$

The standard metric units on electric field strength arise from its definition. Since electric field is defined as a force per charge, its units would be force units divided by charge units. In this case, the standard metric units are Newton/Coulomb or N/C.

Electric Field Due to point charge---See Book (Electricity & Magnetism-R.Murugeshan)

Electric field due to a long uniformly charged wire:

Let us consider a long uniformly charged wire having charge λ coulombs per meter .So, the magnitude of the electric field at p due to charge $dq = \lambda dx$ is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

$$\therefore dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{r^2} \dots \dots \dots (1)$$

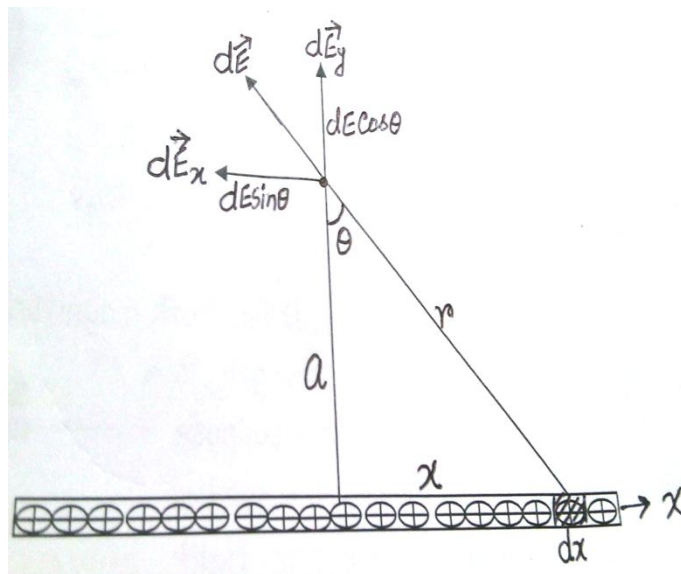


Fig.-A section of an infinite line of charge.

The x component of the resultant vector $\vec{E}$ at p must be zero because every charge element on the right has a corresponding element on the left such that their field contributions in the x -direction cancel. Thus only the y component can contribute to the resultant field at p.

$$dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{r^2} \cos \theta$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{r^2} \frac{a}{r}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \frac{dx}{r^3} \dots \dots \dots (2)$$

Therefore, the electric field at p is,

$$\begin{aligned}
 E_p &= \int_{-\alpha}^{\alpha} dE \cos \theta \\
 &= \int_{-\alpha}^{\alpha} \frac{\lambda a}{4\pi\epsilon_0} \frac{dx}{r^3} \\
 &= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\alpha}^{\alpha} \frac{dx}{(a^2 + x^2)^{3/2}} \dots \dots \dots (3) \quad [\because r = (a^2 + x^2)^{1/2}]
 \end{aligned}$$

Since,

$$\begin{aligned}
 x &= a \tan \theta \\
 \therefore dx &= a \sec^2 \theta d\theta
 \end{aligned}$$

When, $x = \infty$, then $\theta = \frac{\pi}{2}$

When, $x = -\infty$, then $\theta = -\frac{\pi}{2}$

Hence, equation (3) becomes,

$$\begin{aligned}
 E_p &= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{a \sec^2 \theta d\theta}{(a^2 + a^2 \tan^2 \theta)^{3/2}} \\
 &= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{a \sec^2 \theta d\theta}{a^3 \sec^3 \theta} \\
 &= \frac{\lambda}{4\pi\epsilon_0 a} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta d\theta \\
 &= \frac{\lambda}{4\pi\epsilon_0 a} [\sin \theta]_{-\pi/2}^{\pi/2} \\
 &= \frac{\lambda}{4\pi\epsilon_0 a} \left[\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right] \\
 &= \frac{\lambda}{4\pi\epsilon_0 a} [1 - (-1)] \\
 &= \frac{\lambda}{4\pi\epsilon_0 a} \cdot 2 \\
 \therefore E_p &= \frac{\lambda}{2\pi\epsilon_0 a} \dots \dots \dots (4)
 \end{aligned}$$

This is the expression for required electric field.

[From equation (4) it is evident that in the case of a long uniformly charged wire the electric field is inversely proportional to the first power of the distance from the wire ,i.e.,

$$E \propto \frac{1}{a}$$

Where $\frac{\lambda}{2\pi\epsilon_0} = \text{constant.}$]

Electric field at a point on the axis of a charged circular turn of wire:

Let us consider a circular turn of wire of radius a charged with λ coul/m. The electric field at P due to the element of charge $dq = \lambda(a d\phi)$ is,

$$\begin{aligned} dE &= \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \\ &= \frac{\lambda a}{4\pi\epsilon_0} \cdot \frac{d\phi}{r^2} \dots \dots \dots (1) \end{aligned}$$

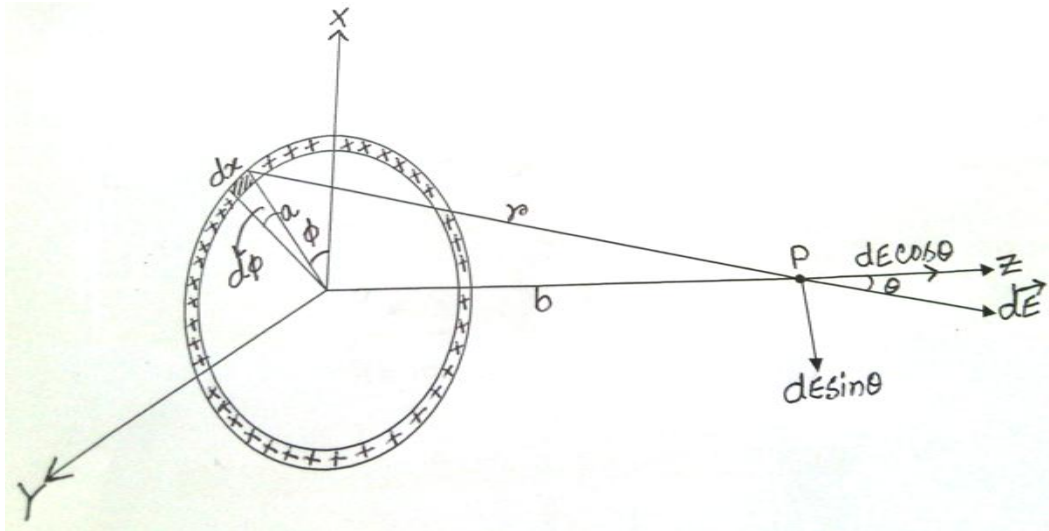


Fig.-Electric field at a point on the axis of a circular turn of charged wire.

From symmetry, the resultant field must lie along the ring axis. Thus only the component of $\vec{dE}$ parallel to this axis contributes to the final result. Therefore, the component,

$$\begin{aligned} dE \cos\theta &= \frac{\lambda a d\phi}{4\pi\epsilon_0 r^2} \cos\theta \\ &= \frac{\lambda a d\phi}{4\pi\epsilon_0 r^2} \frac{b}{r} \\ &= \frac{\lambda a b}{4\pi\epsilon_0 r^3} d\phi \dots \dots \dots (2) \end{aligned}$$

Hence the electric field at P is,

$$\begin{aligned} E_P &= \int_0^{2\pi} dE \cos\theta \\ &= \int_0^{2\pi} \frac{\lambda a b}{4\pi\epsilon_0 r^3} d\phi \\ &= \frac{\lambda a b}{4\pi\epsilon_0 (a^2 + b^2)^{3/2}} [\phi]_0^{2\pi} \\ &= \frac{\lambda a b}{4\pi\epsilon_0 (a^2 + b^2)^{3/2}} 2\pi \\ &= \frac{\lambda a b}{2\epsilon_0 (a^2 + b^2)^{3/2}} \dots \dots \dots (3) \end{aligned}$$

If the total charge on the ring is $q = 2\pi a\lambda$, then

$$E_P = \frac{qb}{4\pi\epsilon_0(a^2 + b^2)^{\frac{3}{2}}} \dots\dots\dots (4)$$

If $b \gg a$, then neglecting a^2 compared with b^2

$$E_P = \frac{q}{4\pi\epsilon_0 b^2} \dots\dots\dots (5)$$

This is the desired expression.

Electric Potential:

The electric potential at a point in an electric field is defined as the work done in bringing a unit positive charge from infinity to that point in question. If W be the work done in bringing a positive charge from infinity to the point, then the electric potential,

$$V = \frac{W}{q_o}.$$

Electric Potential Energy:

If we increase the distance between two charges, an external agent must do work that will be positive if the charges are opposite in sign and negative otherwise. The energy represented by this work can be thought stored in the system as electric potential energy. Considering two charges q_1 and q_2 a distance r apart, the electric potential energy of the system is,

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}.$$

Expression for Electric Potential

Let us consider a test charge q_0 is moved from point A to B along the path AB. The field at the point C at a distance $\vec{r}$ from the origin O due to a source charge q at the origin is,

$$\vec{E} = \frac{q}{4\pi\epsilon_0} \frac{\vec{r}}{r^3} \dots\dots\dots (1)$$

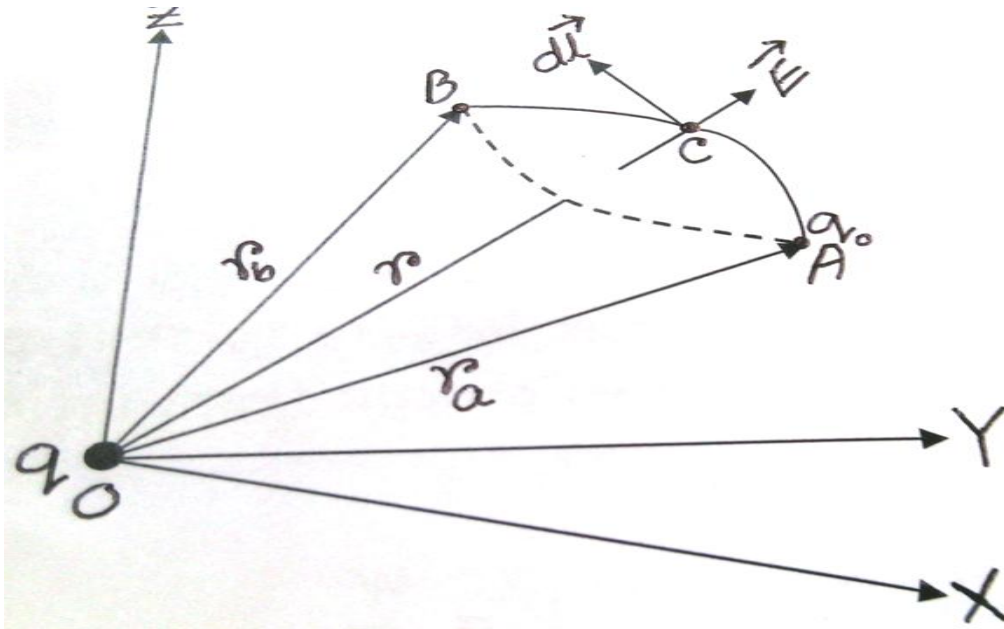


Fig.-01: The path along which the test charge is taken.

Let the elementary distance through which the test charge q_0 at c has been displaced be $d\vec{l}$. Then the elementary work done against the conservative electrostatic force is,

$$\begin{aligned} dW &= -\vec{F} \cdot d\vec{l} \\ &= -q_0 \vec{E} \cdot d\vec{l} \\ &= -\frac{qq_0}{4\pi\epsilon_0} \frac{1}{r^3} \vec{r} \cdot d\vec{l} \dots\dots\dots (2) \text{ [By equation (1)]} \end{aligned}$$

But $\vec{r} \cdot d\vec{l} = r dr$. Hence from (2)

$$\begin{aligned} dW &= -\frac{qq_0}{4\pi\epsilon_0} \frac{r dr}{r^3} \\ &= -\frac{qq_0}{4\pi\epsilon_0} \frac{1}{r^2} dr \dots\dots\dots (3) \end{aligned}$$

Now, the elementary potential difference is,

$$dV = \frac{dW}{q_0} = -\vec{E} \cdot d\vec{l} = -\frac{q}{4\pi\epsilon_0 r^2} dr \dots\dots\dots (4)$$

Therefore, the potential difference between the point B and the point A is,

$$\int_A^B dV = - \int_A^B \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} dr$$

or $V_B - V_A = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_A^B$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right] \dots\dots\dots (5)$$

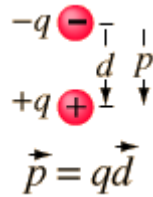
Let the point A be at infinity, then $V_A = 0$ and $\frac{1}{r_A} = 0$. Denoting r_B by r and V_B by V, equation (5) yields,

$$V = \frac{q}{4\pi\epsilon_0 r} \dots\dots\dots (6)$$

which is the required expression.

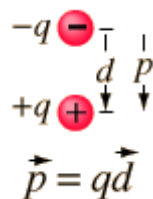
Electric dipole:

When two equal and opposite charges placed at a small distance apart, then the arrangement is called an electric dipole.

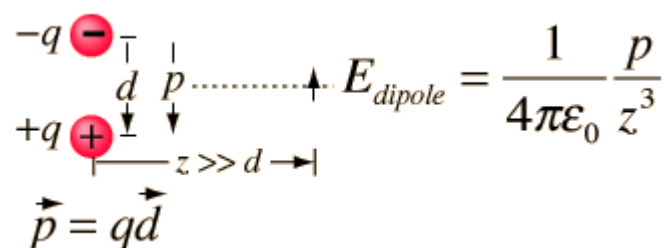


Electric Dipole Moment:

The electric dipole moment for a pair of opposite charges of magnitude q is defined as the magnitude of the charge times the distance between them and the defined direction is toward the positive charge. It is a useful concept in atoms and molecules where the effects of charge separation are measurable, but the distances between the charges are too small to be easily measurable. It is also a useful concept in dielectrics and other applications in solid and liquid materials.



Applications involve the electric field of a dipole and the energy of a dipole when placed in an electric field.



POTENTIAL AND FIELD DUE TO AN ELECTRIC DIPOLE:

Let us consider an electric dipole formed by the charges $-q$ and $+q$ separated by a distance $2l$ along X-axis such that positive X-axis is from charge $-q$ towards charge $+q$ (Fig.-01).

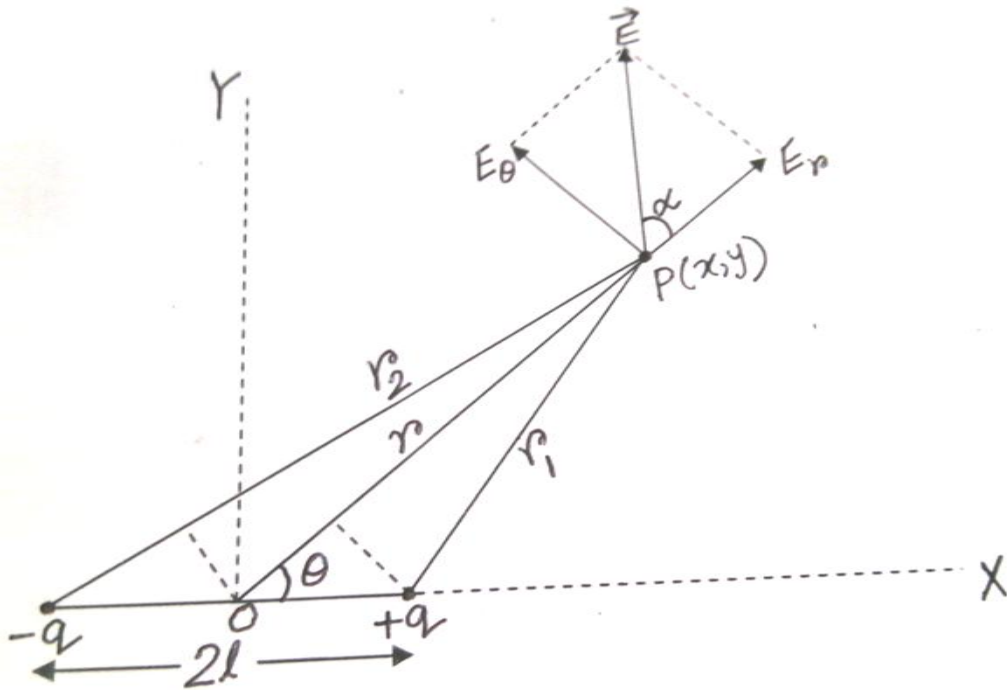


Fig.-01

Let the middle point O of the dipole be the origin and P(x, y) the point at which the potential and field are to be evaluated. Let r be the distance of point P from the origin and θ be the angle which OP makes with the dipole axis.

Obviously the polar coordinates of P are (r, θ) and electric dipole moment $\vec{P} = 2ql\hat{i}$ where $\hat{i}$ is the unit vector along X-axis. If r_1 and r_2 are the distance of point P from the charges $+q$ and $-q$ respectively, the net electric potential V due to the dipole is given by

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r_1} - \frac{q}{r_2} \right] \dots \dots \dots (1)$$

If $r \gg l$, then r_1 and r_2 may be written as

$$\begin{aligned} r_1 &= r - l \cos \theta \\ r_2 &= r + l \cos \theta \end{aligned}$$

Substituting these values in eqn. (1); we get

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r - l \cos \theta} - \frac{q}{r + l \cos \theta} \right]$$

$$\begin{aligned}
&= \frac{1}{4\pi\epsilon_0} \left[\frac{q \cdot 2l \cos \theta}{r^2 - l^2 \cos \theta} \right] \\
&= \frac{1}{4\pi\epsilon_0} \frac{2ql \cos \theta}{r^2} \\
&= \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \dots \dots \dots (2)
\end{aligned}$$

The electric field to P due to the dipole may now be evaluated .It is convenient to use polar co-ordinates (r, θ) in place of Cartesian co-ordinates (x, y) for evaluating the components E_r and E_θ of the electric field are given by

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r} \left[\frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \right] = \frac{1}{4\pi\epsilon_0} \cdot \frac{2p \cos \theta}{r^3} \dots \dots \dots (3)$$

$$E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = -\frac{1}{r} \frac{\partial}{\partial \theta} \left[\frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \right] = \left[\frac{1}{4\pi\epsilon_0} \cdot \frac{p \sin \theta}{r^3} \right] \dots \dots \dots (4)$$

The magnitude of total electric field is given by

$$\begin{aligned}
E &= \sqrt{(E_r^2 + E_\theta^2)} = \frac{1}{4\pi\epsilon_0} \sqrt{\left[\left(\frac{2p \cos \theta}{r^3} \right)^2 + \left(\frac{p \sin \theta}{r^3} \right)^2 \right]} \\
&= \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3} \sqrt{(1 + 3\cos^2 \theta)} \dots \dots \dots (5)
\end{aligned}$$

The angle α which the electric field vector $\vec{E}$ makes with radius vector $\vec{OP}$ is given by

$$\alpha = \tan^{-1} \left(\frac{E_\theta}{E_r} \right) = \tan^{-1} \left(\frac{1}{2} \tan \theta \right)$$

It is obvious from equations (2), (4) and (5) that field and potential of a dipole vary as $\frac{1}{r^3}$ and respectively contrary to the laws of a single charge where the electric field and potential vary as $\frac{1}{r^2}$ and $\frac{1}{r}$ respectively.

Gauss's law and its applications-- See Book (Electricity & Magnetism-R.Murugesan)

Dielectrics-- See Book (Electricity & Magnetism-R.Murugesan)

Gauss's law for dielectrics- See Book (Electricity & Magnetism-R.Murugesan)

Relevant Problem- See Book (Electricity & Magnetism-R.Murugesan)